

Riemann Surfaces

Dibendu Pal

Supervisor: Prof. Dr. Sushil Gorai

IISER-K

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Part-I Overview

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Definitions

Topological manifold

A **topological manifold** X of real dimension $= k$ is a topological Hausdorff space X such that each point $x \in X$ has an open neighbourhood U with a homeomorphism,

$$\phi : U \xrightarrow{\simeq} V$$

onto an open set $V \subset \mathbb{R}^k$.

Complex Atlas

A **complex atlas** of a topological manifold X of real dimension 2, i.e. complex dimension $= 1$, is a family $\mathcal{A}$ of **complex charts**

$$\mathcal{A} = (\phi_i : U_i \rightarrow V_i)_{i \in I}$$

with open subsets

$$V_i \subset \mathbb{C} \simeq \mathbb{R}^2,$$

Complex Atlas

s.t

$$X = \bigcup_{i \in I} U_i$$

and for all pairs $i, j \in I$ and $U_{ij} := U_i \cap U_j$ the transition function of the two charts

$$\psi_{ij} := \phi_i \circ (\phi_j)^{-1} : \phi_j(U_{ij}) \rightarrow \phi_i(U_{ij})$$

is biholomorphic.

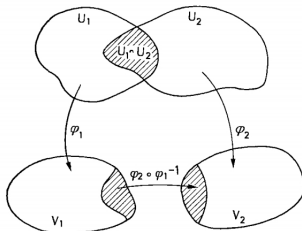


Figure 1

Complex Structure

Two complex atlases $\mathcal{A}_1$ and $\mathcal{A}_2$ of X are biholomorphically compatible if their union $\mathcal{A}_1 \cup \mathcal{A}_2$ is again a complex atlas. A maximal set of complex, biholomorphically compatible atlases of X is a **complex structure** Σ on X .

Riemann Surface

A **Riemann surface** is a pair (X, Σ) with a 2-dimensional connected, topological manifold X with second-countable topology, i.e. having a countable base of the topology, and a complex structure Σ on X .

Holomorphic Map

A continuous map

$$f : (X, \Sigma_X) \rightarrow (Y, \Sigma_Y)$$

between two Riemann surfaces is a **holomorphic map** if for each point $x \in X \exists$ a chart around x from an atlas of Σ_X

$$\phi : U_1 \rightarrow V_1,$$

Holomorphic Map

and a chart around $f(x)$ from an atlas of Σ_Y

$$\psi : U_2 \rightarrow V_2,$$

such that the composition

$$\psi \circ f \circ (\phi)^{-1} : \phi(U_1 \cap f^{-1}(U_2)) \rightarrow V_2$$

is holomorphic in usual sense. A mapping $f : X \rightarrow Y$ is called biholomorphic if it is bijective and both maps $f : X \rightarrow Y$ and $f^{-1} : Y \rightarrow X$ are holomorphic. We note that the definition is independent from the choice of the charts. Also for an open set $Y \subset X$: A map f on Y is holomorphic iff the restriction of f to each component of Y is holomorphic.

Holomorphic Function

A **holomorphic function** on X is a holomorphic map

$$f : (X, \Sigma_X) \rightarrow \mathbb{C}$$

The ring of all holomorphic functions on X with respect to addition and multiplication is denoted $\mathcal{O}(X)$ (also $\mathbb{C}$ Algebra).

Meromorphic Function

A **meromorphic function** on X is given by an open set $U \subset X$ with $X \setminus U$ discrete and a holomorphic function

$$f : U \rightarrow \mathbb{C}$$

such that for all $x_0 \in X \setminus U$

$$\lim_{\substack{x \rightarrow x_0 \\ x \neq x_0}} |f(x)| = \infty.$$

Meromorphic function

The points of $X \setminus U$ are named the **poles** where U is the domain of the meromorphic function. The **order** of a pole $x \in X \setminus U$ is determined by the Laurent expansion of f with respect to a chart around x . The order is also independent of the choice of the chart.

Two meromorphic functions f and g on X can be added and multiplied at each point which is neither a pole of f nor a pole of g . Extending the result to possibly removable singularities defines a meromorphic function on X . The ring of all meromorphic functions on X is denoted $\mathcal{M}(X)$.

For an open set $Y \subset X$: A function f on Y is meromorphic iff the restriction of f to each component of Y is meromorphic.

Examples of Riemann Surfaces

(1) The Complex Plane $\mathbb{C}$.

Its complex structure is defined by the structure whose only chart is the identity map $\mathbb{C} \rightarrow \mathbb{C}$.

(2) Domains .

Assume X is a Riemann surface and $Y \subset X$ is a domain, i.e., a open connected subset. Then Y is also a Riemann surface where its structure includes all those complex charts $\varphi : U \rightarrow V$ on X , where $U \subset Y$. In particular, every domain $Y \subset \mathbb{C}$ is a Riemann surface.

(3) The Riemann sphere $\mathbb{P}^1$.

Let $\mathbb{P}^1 := \mathbb{C} \cup \{\infty\}$, where ∞ is a symbol not contained in $\mathbb{C}$. We introduce the following topology on $\mathbb{P}^1$. The open sets are the usual open sets $U \subset \mathbb{C}$ together with sets of the form $V \cup \{\infty\}$, where $V \subset \mathbb{C}$ is the complement of a compact set $K \subset \mathbb{C}$. With this topology $\mathbb{P}^1$ is a compact Hausdorff topological space, homeomorphic to the 2-sphere S^2 .

Set

$$U_1 := \mathbb{P}^1 \setminus \{\infty\} = \mathbb{C}$$

$$U_2 := \mathbb{P}^1 \setminus \{0\} = \mathbb{C}^* \cup \{\infty\}.$$

Define maps $\varphi_i : U_i \rightarrow \mathbb{C}, i = 1, 2$, as follows:

$$\varphi_1(z) = Id$$

$$\varphi_2(z) := \begin{cases} 1/z & \text{for } z \in \mathbb{C}^* \\ 0 & \text{for } z = \infty \end{cases}$$

These maps are homeomorphisms and so $\mathbb{P}^1$ is a two-dimensional manifold. Since U_1 and U_2 are connected and have non-empty intersection. Hence $U_1 \cup U_2 = \mathbb{P}^1$ is also connected.

(4) Tori

Suppose $\omega_1, \omega_2 \in \mathbb{C}$ are linearly independent over $\mathbb{R}$. Define the lattice

$$\Gamma := \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 = \{n\omega_1 + m\omega_2 : n, m \in \mathbb{Z}\}.$$

Two complex numbers $z, z' \in \mathbb{C}$ are called equivalent mod Γ if $z - z' \in \Gamma$. The set of all equivalence classes is denoted by $\mathbb{C}/\Gamma$. Let $\pi : \mathbb{C} \rightarrow \mathbb{C}/\Gamma$ be the natural projection, i.e., the map that associates to each point $z \in \mathbb{C}$ its equivalence class mod Γ .

We define the quotient topology on $\mathbb{C}/\Gamma$. A subset $U \subset \mathbb{C}/\Gamma$ is open precisely if $\pi^{-1}(U) \subset \mathbb{C}$ is open. With this topology $\mathbb{C}/\Gamma$ is a Hausdorff topological space and the quotient map $\pi : \mathbb{C} \rightarrow \mathbb{C}/\Gamma$ is continuous. Since $\mathbb{C}$ is connected, $\mathbb{C}/\Gamma$ is also connected (as connectedness is preserved under continuous map). Also $\mathbb{C}/\Gamma$ is compact, as $\pi(P) = \mathbb{C}/\Gamma$ where,

$$P := \{\lambda\omega_1 + \mu\omega_2 : \lambda, \mu \in [0, 1]\}.$$

The map π is also open, i.e., the image of every open set $V \subset \mathbb{C}$ is open. Since,

$$\pi^{-1}(\pi(V)) = \hat{V} = \bigcup_{\omega \in \Gamma} (\omega + V).$$

and every set $\omega + V$ is open, so $\hat{V}$ is open.

The complex structure on $\mathbb{C}/\Gamma$ is defined in the following way. Let $V \subset \mathbb{C}$ be an open set such that no two points in V are equivalent under Γ . Then $U := \pi(V)$ is open and $\pi|_V : V \rightarrow U$ is a homeomorphism (continuous, open, injective and surjective). Its inverse $\varphi = (\pi|_V)^{-1} : U \rightarrow V$ is a complex chart on $\mathbb{C}/\Gamma$. Let the complex structure be the set of all charts obtained in this way. We have to show that any two charts $\varphi_i : U_i \rightarrow V_i, i = 1, 2$, are holomorphically compatible. Consider the map

$$\psi := \varphi_2 \circ \varphi_1^{-1} : \varphi_1(U_1 \cap U_2) \rightarrow \varphi_2(U_1 \cap U_2).$$

For every $z \in \varphi_1(U_1 \cap U_2)$ we have $\pi(\psi(z)) = \pi(\varphi_2 \circ (\varphi_1)^{-1})(z) = \varphi_1^{-1}(z) = \pi(z)$ and thus $\psi(z) - z \in \Gamma$. Since Γ is discrete and ψ is continuous, this implies that $\psi(z) - z$ is constant on every connected component of $\varphi_1(U_1 \cap U_2)$. Thus ψ is holomorphic. Similarly ψ^{-1} is also holomorphic.

Some Theorems .

Riemann's Removable Singularity Theorem

Let U be an open subset of Riemann surface X and let $a \in U$. Suppose the holomorphic function $f \in \mathcal{O}(U \setminus \{a\})$ is bounded in some nbd of a . Then f can be extended uniquely to a function $\tilde{f} \in \mathcal{O}(U)$.

Proof: We apply the normal Riemann Removable Singularity theorem on a chart around a inside U where the function $f \circ (\phi)^{-1}$ is holomorphic and bounded in $\phi(U) \setminus \phi(a)$.

Identity Theorem

Suppose X and Y are Riemann surfaces and $f_1, f_2 : X \rightarrow Y$ are two holomorphic mappings s.t. $f_1 = f_2$ on $A \subset X$ having a limit point $a \in X$. Then f_1 and f_2 are identically equal.

Proof: Let $G = \{x \in X | \exists \text{ an open nbd of } x \in W \text{ s.t. } f_1|_W = f_2|_W\}$. By definition G is open.

Claim: G is also closed.

Suppose b is a boundary point of G . Then $f_1(b) = f_2(b)$ since f_1 and f_2 are continuous. Choose charts $\varphi : U \rightarrow V$ on X and $\psi : U' \rightarrow V'$ on Y with $b \in U$ and $f_i(U) \subset U'$. We may also assume that U is connected. Now, the mappings

$$g_i := \psi \circ f_i \circ \varphi^{-1} : V \rightarrow V' \subset \mathbb{C}$$

are holomorphic. Also $U \cap G \neq \emptyset$, and open. Since

$g_1|_{\phi(U \cap G)} = g_2|_{\phi(U \cap G)}$ the Identity Theorem for holomorphic functions on $\mathbb{C}$ implies $g_1 = g_2$ on $\phi(U \cap G)$. Thus $f_1|_U = f_2|_U$. Hence $b \in G$ and thus G is closed. Now since X is connected either $G = \emptyset$ or $G = X$. But the first case is excluded since $a \in G$ (using the Identity Theorem in the plane in the above same way). Hence $f_1 = f_2$ on X .

Theorem (Meromorphic functions and holomorphic maps)

Meromorphic functions on a Riemann surface X are holomorphic maps

$$X \rightarrow \mathbb{P}^1.$$

Conversely, any non-constant holomorphic map

$$f : X \rightarrow \mathbb{P}^1$$

is a meromorphic function $f \in \mathcal{M}(X)$ with domain $X \setminus f^{-1}(\infty)$.

Proof. i) Consider a meromorphic function $f \in \mathcal{M}(X)$. For each pole $x_0 \in X$ extend f by defining

$$f(x_0) := \infty \in \mathbb{P}^1.$$

Using the standard atlas of $\mathbb{P}^1$, for a suitable neighbourhood $U \subset X$ of x_0 the map

$$\phi_0 \circ (f \mid U) : U \rightarrow \mathbb{C}$$

is continuous, holomorphic and bounded on $U \setminus \{x_0\}$. By Riemann's theorem about removable singularities the map

$$\phi_0 \circ (f \mid U)$$

is holomorphic. Hence the extension

$$f : X \rightarrow \mathbb{P}^1$$

is a holomorphic map. ii) Let

$$f : X \rightarrow \mathbb{P}^1$$

be a non-constant holomorphic map. Then if

$$f^{-1}(\infty) \subset X$$

is discrete with

$$f \mid X' : X' \rightarrow \mathbb{C}, X' := X \setminus f^{-1}(\infty),$$

holomorphic and for any $x_0 \in f^{-1}(\infty)$

$$\lim_{x \rightarrow x_0} f(x) = \infty$$

If $f^{-1}(\infty)$ is not discrete then there exists a limit point of the set $f^{-1}(X)$ in X . Since, the map $g(x) = \infty$ is holomorphic from X to $\mathbb{P}^1$ and is equal on the set $f^{-1}(X)$. Then by the identity theorem $f = g$ ($\rightarrow \leftarrow$)

Local Representation of Holomorphic Maps

Theorem (Local representation of a holomorphic map)

Consider a nonconstant holomorphic map

$$f : X \rightarrow Y$$

between two Riemann surfaces. For any $x \in X$ a complex chart of X around x , $\phi : U \rightarrow V$ and a complex chart of Y around $f(x)$, $\psi : S \rightarrow T$, such that

$$f(U) \subset S$$

and

$$g := \psi \circ f \circ \phi^{-1} : V \rightarrow T$$

has the form

$$g(z) = z^k, z \in V.$$

Proof. i) Choosing charts: We choose charts

$$\phi_1 : U_1 \rightarrow V_1$$

of X around x and

$$\psi : S \rightarrow T$$

of Y around $f(x)$ with

$$\phi_1(x) = \psi(f(x)) = 0 \in \mathbb{C}$$

and

$$f(U_1) \subset S.$$

Then

$$g_1 := \psi \circ f \circ \phi_1^{-1} : V_1 \rightarrow T$$

is a non-constant holomorphic function with

$$g_1(0) = 0.$$

If

$$k := \text{ord}(g_1; 0) \geq 1$$

then

$$g_1(w) = w^k \cdot h_1(w)$$

with h_1 holomorphic, having no zeros in a neighbourhood $D_r(0)$.

ii) Existence of a k -th root: Now the function h_1 has a k -th root

$$h := \sqrt[k]{h_1} : D_r(0) \rightarrow \mathbb{C}^*$$

Since, $h(0) \neq 0$ the function

$$D_r(0) \rightarrow \mathbb{C}, w \mapsto w \cdot h(w)$$

is locally biholomorphic in a neighbourhood of zero by the inverse mapping theorem. Hence for suitable neighbourhoods of 0

$$V, V_2 \subset V_1$$

its restriction

$$\alpha : V_2 \xrightarrow{\sim} V, w \mapsto w \cdot h(w), \text{ is biholomorphic}$$

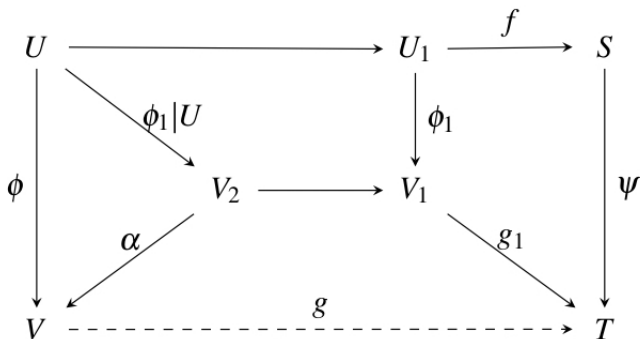
. iii) The chart needed: Elements $z \in V$ satisfy

$$z = \alpha(w) = w \cdot h(w) \text{ or } \alpha^{-1}(z) = w$$

For $U := \phi_1^{-1}(V_2)$ the map

$$\phi := \alpha \circ (\phi_1|_U) : U \rightarrow V$$

is a complex chart of X around x .



Define

$$g := g_1 \circ \alpha^{-1} = \psi \circ f \circ \phi^{-1}$$

For

$$z = \alpha(w) = w \cdot h(w) \in V$$

then

$$g(z) = g_1(\alpha^{-1}(z)) = g_1(w) = (w \cdot h(w))^k = z^k,$$

Open Mapping Theorem

Corollary . Each non-constant holomorphic map

$$f : X \rightarrow Y$$

between two Riemann surfaces is an open map.

Proof. Last theorem implies that f maps U a neighbourhood of a point $x \in X$ to neighbourhood $zf(U)$ of $f(x)$ in Y .

Corollary

Any injective holomorphic map between Riemann surfaces

$$f : X \rightarrow Y$$

is biholomorphic onto its image, $f(X)$

Proof: Since, f is injective the local description in last theorem gives $k=1$. Hence the inverse mapping $f^{-1} : f(X) \rightarrow X$ is holomorphic.

Theorem

Suppose X and Y are Riemann surfaces. Suppose X is compact and $f : X \rightarrow Y$ is a non-constant holomorphic mapping. Then Y is compact and f is surjective.

Proof. $f(X)$ is open by open mapping theorem. Since X is compact, $f(X)$ is compact and thus closed. Since the only subsets of a connected topological space which are both open and closed are the empty set and the whole space, it follows that $f(X) = Y$. Thus f is surjective and Y compact.

Corollary.

Every holomorphic function on a compact Riemann surface is constant.